Comparative Error Analysis of POS Taggers on UD English EWT

Comparing classical and neural POS taggers through results, trade-offs, and error patterns

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Compact BiLSTM achieves the best raw quality, CRF is the strongest practical trade-off, and PROPN ↔ NOUN remains the main unresolved ambiguity across all models.

1 Motivation & Research Question

We compare classical and neural POS taggers on the same benchmark to understand not only which model performs best, but also why models fail and which model family is most useful in practice.

- shared UD English EWT benchmark and UPOS label inventory
- compare quality, efficiency, robustness, and interpretability
- analyze confusions, long-sentence failures, and OOV sensitivity

Which POS tagging model gives the best balance between final quality, robustness, efficiency, and interpretability?

2 Dataset, Models & Setup

Split	Sentences	Tokens	Avg len	Vocab	Token OOV
Train	12,544	204,577	16.309	19,674	-
Dev	2,001	25,147	12.567	5,494	0.08303
Test	2,077	25,094	12.082	5,629	0.09134

Sentence-level OOV: Dev 0.12843, Test 0.13837

Averaged Perceptron	strong feature-based baseline, simple and interpretable
HMM + Viterbi	tiny probabilistic sequence baseline, highly interpretable
CRF	strongest classical structured model, balances quality and interpretability
Compact BiLSTM	neural tagger with word and character information, strongest final quality

Load benchmark → build vocabulary/ labels → train each model → evaluate on dev/test → analyze errors

3 Main Results

Model	Accuracy	Macro-F1	Train time (s)	Inference (s)	Tokens/s	Size (MB)
Compact BiLSTM	0.952419	0.904233	574.5724	1.2011	20,891.80	11.9991
CRF	0.946880	0.899161	22.9582	0.1398	179,476.44	4.6725
Averaged Perceptron	0.943652	0.894724	32.7098	0.6227	40,296.88	14.6633
HMM + Viterbi	0.911294	0.840027	0.1010	2.8858	8,695.58	0.6275

Best raw quality: Compact BiLSTM

Best practical trade-off: CRF

Smallest / fastest to train: HMM + Viterbi

4 What the Results Mean

A BiLSTM vs CRF

- +0.554 pp accuracy
- +0.507 pp macro-F1
- 25.0× higher training cost
- 8.6× slower inference
- 2.57× larger model

The neural model is best by score, but the gain over CRF is modest relative to its much higher computational cost.

B CRF vs Averaged Perceptron

- +0.323 pp accuracy
- +0.444 pp macro-F1
- about 4.45× higher throughput
- about 3.1× smaller model

CRF is not just slightly better — in this setup it improves both quality and operational efficiency.

C Non-obvious takeaway

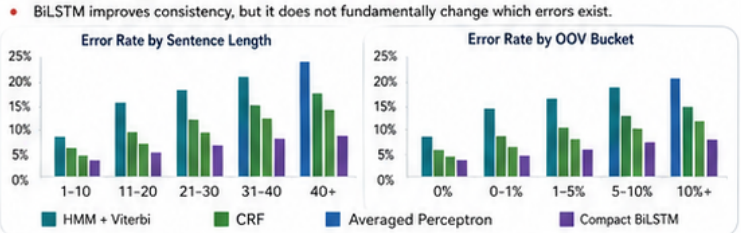
Model quality does not follow a simple bigger-is-better pattern. CRF is much smaller than BiLSTM yet very close in quality, which makes the model-selection story more meaningful than a simple leaderboard.

5 Error Analysis & Robustness

The main unresolved ambiguity across all models is PROPN ↔ NOUN.

Model	PROPN→NOUN	NOUN→PROPN	Other top error
Averaged Perceptron	224	184	VERB→NOUN 78
HMM + Viterbi	375	161	VERB→ AUX 105
CRF	210	174	VERB→NOUN 70
Compact BiLSTM	218	158	VERB→NOUN 59

- Proper nouns depend strongly on context, casing, and lexical familiarity.
- HMM shows a distinctive VERB → AUX pattern, reflecting rigid transition/ emission statistics.
- BiLSTM improves consistency, but it does not fundamentally change which errors exist.



Hard cases cluster in long, OOV-heavy, name-rich sentences.

Easiest tags: PUNCT, CCONJ, DET, PRON, AUX. **Hardest tags:** X, SCONJ, PROPN, SYM, INTJ.

6 Conclusions & Recommendation

The best model depends on the constraint.

Best raw quality — Compact BILSTM
highest accuracy and macro-F1

Best practical trade-off — CRF
near-best quality, much faster, cheaper, and interpretable

Best simple baseline — Averaged Perceptron
strong results with simple design

Best pedagogical baseline — HMM + Viterbi
smallest model, fastest training, easiest probabilistic interpretation

The best model by score is not the best model by practicality. Neural tagging gives the best raw score, but CRF is the most balanced recommendation when quality, speed, model size, and interpretability all matter.

UD English EWT • POS tagging • error analysis • sequence labeling • CRF • BiLSTM